How International Players Have Transformed the NBA
Brandon Yau | DSCI 510 Final Project | May 10, 2026

Motivation for the project

The NBA used to be an almost entirely American league. That story has shifted dramatically in the last decade. The names tell most of it: Giannis Antetokounmpo, Nikola Jokić, Luka Dončić, and Shai Gilgeous-Alexander are running the league today, and the MVP award has gone to an internationally-born player in six of the last seven seasons (all seven if you count Joel Embiid, but Wikipedia lists him as American). As an avid basketball fan, this is the most interesting demographic shift in the sport, and it raises real analytical questions: which countries are producing these players, are they being valued correctly in the draft, and is the league actually playing differently as their share of minutes has grown? The figures produced from this project are designed to answer these questions.

Source 1 - Wikipedia: List of NBA players born outside the United States

URL: https://en.wikipedia.org/wiki/List_of_NBA_players_born_outside_the_United_States

This page contains a single sortable table where every row is a player born outside the U.S. who has appeared in an NBA game, along with their nationality. I scraped it with the requests library and parsed it with BeautifulSoup, locating the table by checking for the presence of the "Nationality" and "Player" column headers, then extracting the player name from the anchor tag inside each player cell. The scrape returns 632 unique (player, country) pairs across 83 countries.

Source 2 - Basketball-Reference: NBA draft history

URL pattern: https://www.basketball-reference.com/leagues/NBA_{year}_advanced.html

Basketball-Reference posts a table of every drafted player for each season, including pick number, team, and aggregate career statistics. I pulled ten draft pages (one per year from 2015 to 2024) using pandas.read_html(), and kept the columns I needed: pick number, player name, career Win Shares (WS), and career Value Over Replacement Player (VORP). Total: 604 picks.

Source 3 - Basketball-Reference: Per-season advanced stats

URL pattern: https://www.basketball-reference.com/leagues/NBA_{year}_advanced.html

Each of these pages contains a row for every player who appeared in the NBA that season, with rate-based metrics like Player Efficiency Rating (PER), True Shooting Percentage (TS%), Usage Percentage (USG%), Box Plus/Minus (BPM), and VORP. It is important to note that this VORP is single-season, unlike the career VORP pulled from the draft history pages. I extracted ten seasons of these tables using pandas.read_html(). The result is 6,789 player seasons, which is the basis for any analysis that asks "how did this player do this year" rather than "how has this player done overall."

Changes from the original proposal

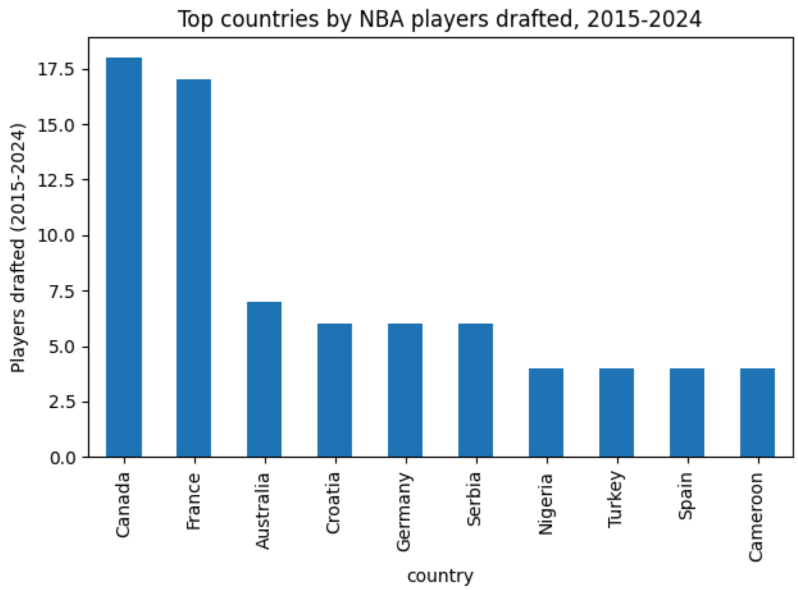
My proposal specified the nba_api Python library for Source 3 instead of Basketball-Reference. When I ran it on my laptop, every request to stats.nba.com returned a JSONDecodeError, because the NBA blocks datacenter and many residential IP ranges from hitting their stats backend. To work around this, I switched to scraping the equivalent advanced stats tables on Basketball-Reference. The schema and the metrics I needed are the same, so the analysis was unaffected.

Integrated data model

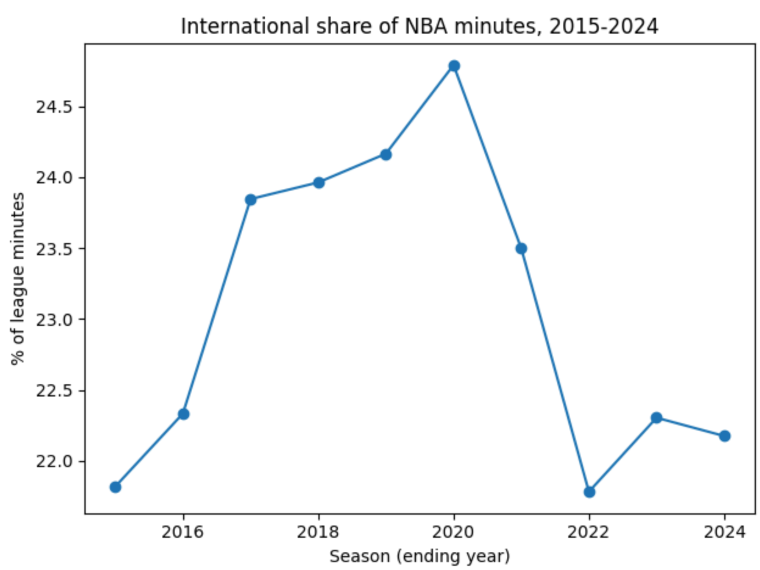
All three sources are joined on a common key labeled player_key, which is a normalized version of the player's name. It was made lowercase, ASCII, no punctuation, and no suffixes like Jr. or III. The ASCII part is particularly important due to much of the international playerbase having names with diacritics, such as Luka Dončić. Without it, many names would not match and Wikipedia includes these diacritics while Basketball-Reference does not. The ER diagram is as attached:

Two integrated tables are produced. players_integrated.csv joins the draft data to Wikipedia and adds an is_international flag (true if the player appears in the Wikipedia list, false otherwise). Seasons_integrated.csv does the same for the advanced stats. Of 594 picks made between 2015 and 2024, 128 match the Wikipedia international list.

Which countries are producing the most players?

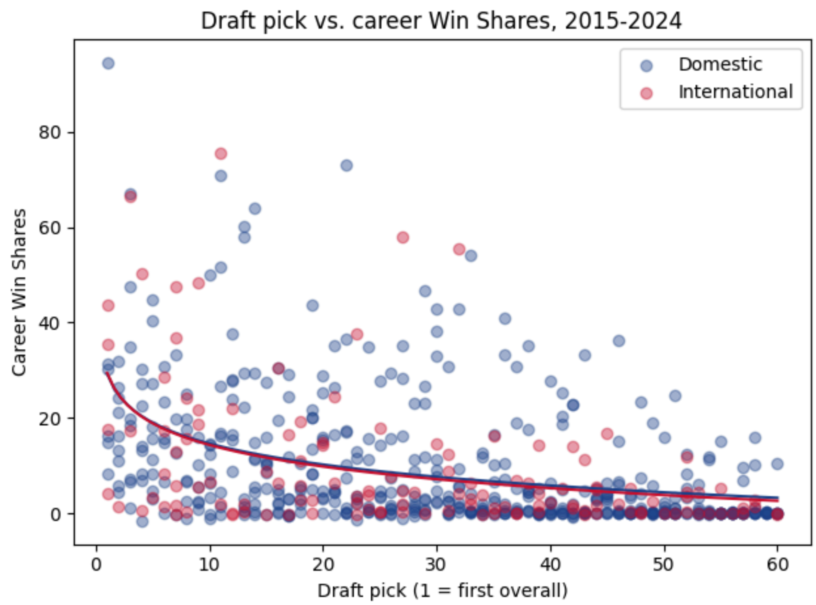


It appears that two countries dominate the international pipeline: Canada with 18 picks and France at 17. After that there is a steep drop-off to Australia (7), then a tier of European countries with six picks each (Croatia, Germany, Serbia). The top-2 dominance is large enough that any "globalization" narrative for the NBA over this decade can be considered a Canada-and-France story, with Canadian Shai Gilgeous-Alexander being the reigning MVP and champion and Frenchman Victor Wembanyama's rising trajectory after being drafted first overall in 2023.

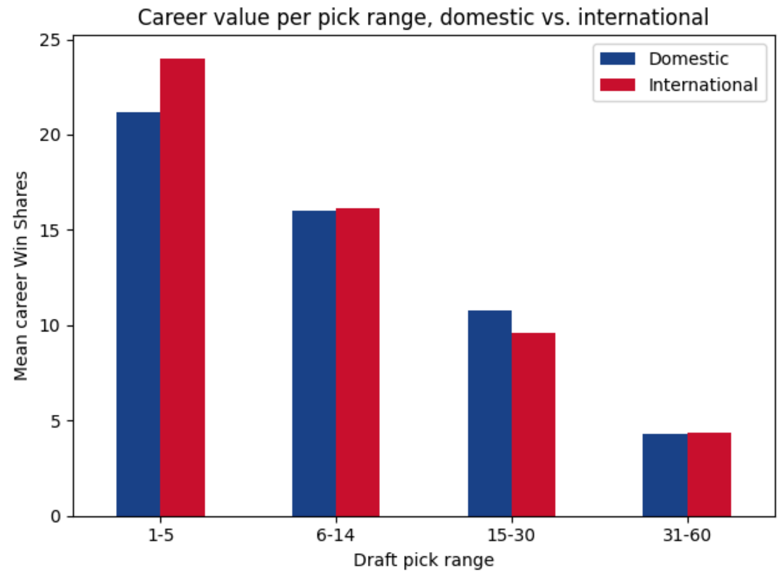


Interestingly enough, the international share of the league was not the explosive growth curve I expected. The graph shows that international share moved within a fairly narrow band of 22% to 25%, with a peak 2020 and a dip afterwards. This shows that it is not a recent flood of new international depth into the league, but rather that we are seeing an explosion of superstars. International players are increasingly becoming the best ones in the league.

Are international players undervalued in the draft?

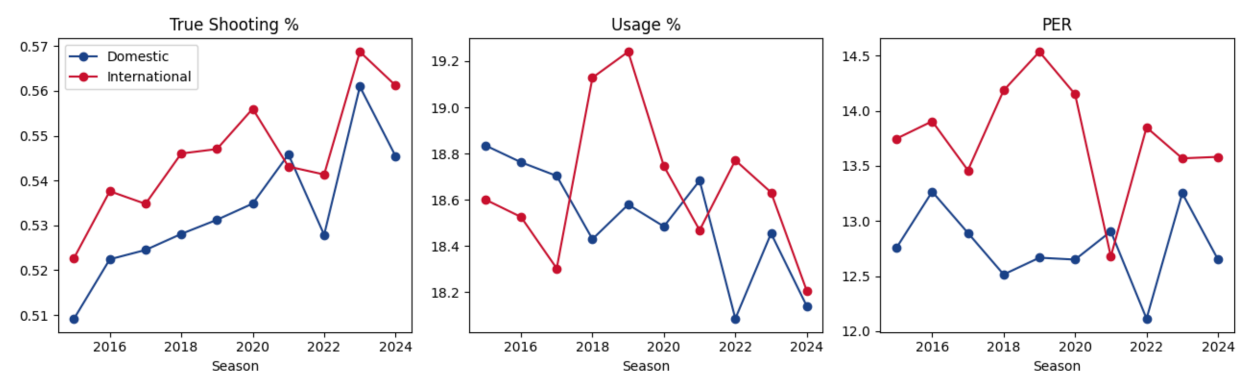


This was the hardest question to answer with numbers. In the end, I decided to represent it with a scatterplot comparing career Win Shares to draft pick number, with colors separating the domestic and international players. Each dot is a single drafted player. To summarize the overall trend, I used a log-linear regression. A plain linear regression would assume that going from pick 1 to pick 2 is worth the same amount of WS as going from pick 50 to pick 51. In basketball, the number 1 pick is dramatically more valuable, while pick 50 and pick 51 are basically interchangeable. With that in mind, the two regression curves are nearly indistinguishable—meaning that, on average, the career value returned by an international pick is about the same as a domestic pick at the same draft position.



This is where the nuances begin to show. As you can see, after pick 6, mean career win shares are similar between international and domestic. However, in the lottery (picks 1-5), international players performed significantly better with 24.0 mean career win shares as compared to 21.2 for domestic. One thing to note is that the sample for international players is much smaller, with only 10 lottery picks. This leads to higher variance and shows us that when an international is picked within the first 5, they are more likely to perform well across their career. This may have to do with the amount of scrutiny that goes into scouting international players, but I won't get into that.

Is the international playing style different?



This was the most striking finding. Across all ten seasons, internationally-born players averaged higher True Shooting %, higher Usage %, and substantially higher PER than domestic players. The PER gap in particular is wide and stable: international players averaged roughly 13.5–14.5 PER each season, while domestic players hovered around 12.5–13. PER is a per-minute composite metric so this isn't a function of more playing time: it means that minute for minute, the average international player produced more across the box score than the average domestic player. This reveals some selection bias: the international population is filtered through the NBA scouting funnel from a much larger pool of players globally, so only the best survive. But the size and consistency of the gap is still notable, and it is consistent with a story in which international players who reach the NBA tend to be skill-and-shooting types who fit cleanly into the modern game.

Conclusions

The data tells a more nuanced story than the popular one. The NBA is not being overrun by international players—their share of minutes has been roughly flat at 22–25% for ten years. What has happened is that the very top of the international talent distribution has gotten extraordinarily good, and is increasingly producing the league's best players: lottery internationals have outperformed lottery domestics by about three Win Shares on average over the decade, and international players have outperformed domestic ones on per-minute efficiency every single season. Two countries account for most of this. Canada and France contributed roughly twice as many drafted players as any other country and are responsible for a large share of the recent international stars. The pipeline from those two countries is now mature enough that it produces multiple lottery-caliber players per year, which was not true a decade ago. On the question of draft valuation: the average international pick at any given slot has returned about the same career value as the average domestic pick, but the variance is wider. The biggest hits and the biggest misses are both more likely to come from the international pool. Whether that constitutes "undervalued" depends on how teams evaluate that variance, which is a question for their front offices, not this dataset.

Future work

The biggest limitation of this analysis is sample size. The international group only has 128 drafted players over the ten-year window, and that gets sliced thinner when bucketed by draft position (only 10 international lottery picks, for example). With samples that small, a few standout careers like Dončić or Wembanyama can substantially move the group averages, so I would not treat the lottery-value gap as conclusive without more data. The most direct extension is to widen the time window to 2005–2024, which would roughly double the international sample and let me distinguish a real era effect from short-term noise. The tradeoff is that pre-2010 picks have far longer careers than recent ones, so the career-WS comparison would need to be restricted to a window where everyone has had at least five years to accumulate stats. Beyond more data, three extensions would make this richer. Adding salary data would let me ask the more interesting valuation question whether international players are outperforming their contracts, assessing current value rather than their pre-NBA draft stock. The analysis also divides players into either international or domestic. A richer analysis would account for naturalized players (Joel Embiid, born in Cameroon, lists U.S. nationality on Wikipedia and is therefore classified as domestic in this dataset) and the difference between players who developed entirely abroad versus those who came through U.S. college basketball.